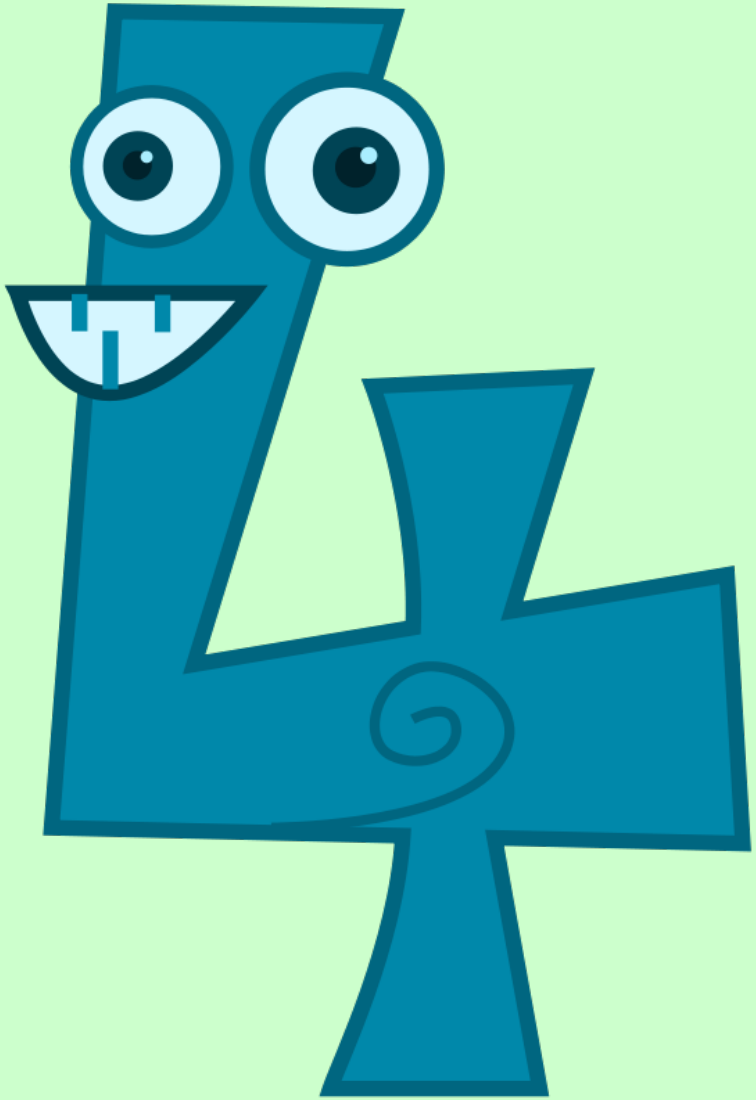


Four Vectors

We live in 3+1 **spacetime** not 3D space. So we need vectors that have four components

We've already seen several of these

1. Events are actually 4-position $x = (ct, x, y, z)$

Up until now we've been scaling x, y, and z to be in light seconds

$x = (t, x, y, z)$ So all coordinates have the same footing

Note how we're doing **seconds of space**

The other option, as we're doing here is to scale time into meters

ct = (3×10^8 m/s)(1s) = 3×10^8 m

1 meter of time then is 1/3e8 = 3.34 ns

Ex. How long in seconds is 1,500,000 meters of time?

2. Spacetime separation are actually 4-displacement

$$\Delta s = (c\Delta t, \Delta x, \Delta y, \Delta z)$$

Four vectors are shown either with **bold** letters or tildes underneath symbols

$$\mathbf{\Delta s} \quad \text{or} \quad \underset{\sim}{\Delta s}$$

The interesting thing about four-vectors is their dot-product is some invariant quantity

Ex. $\mathbf{A} = \begin{bmatrix} A^t \\ A^x \\ A^y \\ A^z \end{bmatrix} = \begin{bmatrix} -5 \\ 3 \\ 1 \\ 2 \end{bmatrix}$

$$\mathbf{A} \cdot \mathbf{A} = -(A^t)^2 + (A^x)^2 + (A^y)^2 + (A^z)^2$$

$$\mathbf{A} \cdot \mathbf{A} = -(-5)^2 + (3)^2 + (1)^2 + (2)^2$$

$$\mathbf{A} \cdot \mathbf{A} = -11$$

It actually works out to – the invariant squared

$$\mathbf{A} \cdot \mathbf{A} = -\left(\sqrt{11}\right)^2$$ The invariant is $\sqrt{11}$

The four-vector variant is the same in any frame, because its invariant

Lets find the dot product of spacetime interval with its self

$$\mathbf{\Delta s} \cdot \mathbf{\Delta s} = \Delta s^2$$

$$\mathbf{\Delta s} \cdot \mathbf{\Delta s} = \begin{bmatrix} c\Delta t \\ \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} \cdot \begin{bmatrix} c\Delta t \\ \Delta x \\ \Delta y \\ \Delta z \end{bmatrix} = -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

We know the invariant here, proper time, so:

$$\Delta s^2 = \mathbf{\Delta s} \cdot \mathbf{\Delta s} = -(c\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2 = -(\tau)^2$$

Assuming no motion in the y or z, we can rearrange

$$\boxed{(c\Delta t)^2 - (\Delta x)^2 = \tau^2}$$

Ex. Event A is at (ct,x) =(3 m,5 m). What is the proper time between events?
Event B is at (ct, x) =(9 m, 7 m).

3. New Quantity: 4-velocity $\mathbf{u} = \frac{ds}{d\tau} \approx \frac{\Delta s}{\Delta \tau} = \begin{bmatrix} c\frac{dt}{d\tau} \\ \frac{dx}{d\tau} \end{bmatrix}$

$$\begin{aligned} t &= \gamma\tau \\ dt &= \gamma d\tau \\ \frac{dt}{d\tau} &= \gamma \end{aligned}$$

$$\frac{dx(t)}{d\tau} = \frac{dx}{d\tau}\frac{dt}{d\tau} = \boxed{v_x\gamma}$$

$$\mathbf{u} = \begin{bmatrix} u^t \\ u^x \\ u^y \\ u^z \end{bmatrix} = \begin{bmatrix} \gamma c \\ \gamma v_x \\ \gamma v_y \\ \gamma v_z \end{bmatrix}$$

Lets take the dot product of **u** with itself

$$\mathbf{u}\cdot\mathbf{u} = -(\gamma c)^2 + (\gamma v_x)^2 = -(\text{Invariant})^2$$

Whatever could the invariant velocity be? c

$$\mathbf{u}\cdot\mathbf{u} = -(\gamma c)^2 + (\gamma v_x)^2 = -(c)^2$$

Can be rearranged and written in terms of components: $\boxed{(u^t)^2 - (u^x)^2 = c^2}$

Ex. If $\beta = 0$ Find ut and ux. Interpret these results

Ex. If $\beta = 3/5$ Find ut and ux. Interpret these results

4. New Quantity: 4-momentum $\mathbf{u} = \begin{bmatrix} \gamma c \\ \gamma v_x \\ \gamma v_y \\ \gamma v_z \end{bmatrix}$ And p = mv

Try: $\text{“}\mathbf{p}\text{”} = m\mathbf{u} = \begin{bmatrix} \gamma mc \\ \gamma mv_x \\ \gamma mv_y \\ \gamma mv_z \end{bmatrix}$

Then $\text{“}\mathbf{p}\text{”} \cdot \text{“}\mathbf{p}\text{”} = m^2\mathbf{u}\cdot\mathbf{u} = -(\gamma mc)^2 + (\gamma mv_x)^2 + \cdots$

Since we already know that $\mathbf{u}\cdot\mathbf{u} = -c^2$

$$-(mc)^2 = -(\gamma mc)^2 + (\gamma mv_x)^2 + \cdots$$

But from Electro-magnetism we know that $\Delta\vec{p} = \frac{\Delta E}{c}$ or $E = pc$

Apart from c, E = p so these need to be unified.
The momentum 4-vector is really $\mathbf{p}c \cdot \mathbf{p}c$ not $\mathbf{p}\cdot\mathbf{p}$, so multiply through by c^2

$$-(mc^2)^2 = -(\gamma mc^2)^2 + (\gamma mcv_x)^2 + \cdots$$

Look at the equation: $-(mc^2)^2 = -(\gamma mc^2)^2 + (\gamma mcv_x)^2 + \cdots$

The invariant Energy $E_0 = mc^2$

Also called rest mass energy

Total Energy $(E)^2$

$\vec{p}_x = \gamma m \vec{v}_x$

So term is: $(p_x c)^2$

Which is an energy

$$-(E_0)^2 = -E^2 + (p_x c)^2$$

So a better way to write the momentum four vector would have been:

$$\mathbf{p} = \begin{bmatrix} E \\ p_x c \\ p_y c \\ p_z c \end{bmatrix}$$

And $\mathbf{p} \cdot \mathbf{p} = -(E_0)^2 = -E^2 + (p_x c)^2 + (p_y c)^2 + (p_z c)^2$

In Sum:

1. Relativistic Momentum $\vec{p}_x = \gamma m \vec{v}_x$
2. Rest Energy $E_0 = mc^2$
3. Total Energy $E = \gamma mc^2$
4. Kinetic Energy $KE = (\gamma - 1)mc^2$

Kinetic energy?

$$\begin{aligned} E &= E_0 + KE \\ \gamma mc^2 &= mc^2 + KE \\ KE &= (\gamma - 1)mc^2 \end{aligned}$$

Ex. Proton traveling at v = 3/5 c in +x direction. $M_p = 1.67 \times 10^{-27}$ kg

A. What is the rest energy, E_0 ?

B. What is the total energy, E ?

C. What is the x-momentum, p_x ?

D. What is the KE?

Any four vector transforms with the Lorentz transformation!

$$\mathbf{p}' = \Lambda \mathbf{P} \qquad \mathbf{p} = \Lambda^{-1} \mathbf{P}'$$

If frame moving +x

If frame moving -x

Assuming all motion in x only:

$$\mathbf{p}' = \Lambda \mathbf{P}$$

$$\begin{bmatrix} E' \\ p_x' c \end{bmatrix} = \begin{bmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{bmatrix} \begin{bmatrix} E \\ p_x c \end{bmatrix}$$

Its so easy to find E and p_x in other frames!!

Ex. Proton traveling at v = 3/5 c in +x direction. $M_p = 1.67 \times 10^{-27}$ kg

Find E and px in S', which is moving to the right at 4/5 c

Ex. A 1g particle has a velocity of 0.8c

A. Find rest Energy
B. Find Total Energy
C. Find Kinetic Energy
D. Find x-momentum, p_x
E. Find $p_x c$
F. Find 4-momentum
G. Show that $E^2 - (pc)^2 = E_0^2$

Another frame moves by going at 0.9c in –x direction relative to the first.

H. Find the components of the 4-momentum in that Frame